BREAST CANCER DETECTION

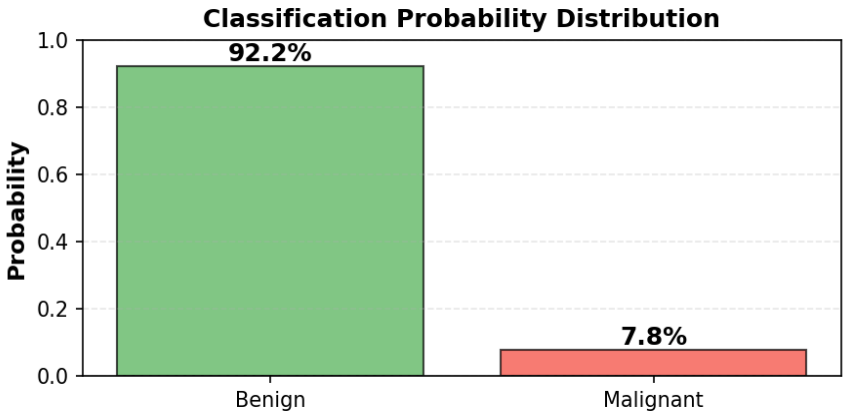
AI-Assisted Histopathology Analysis Report

Report ID:	BCR-20260216_192618
Generated:	2026-02-16 19:26:19
Patient ID:	TEST001
Image File:	benign_0000.png

DIAGNOSTIC RESULTS

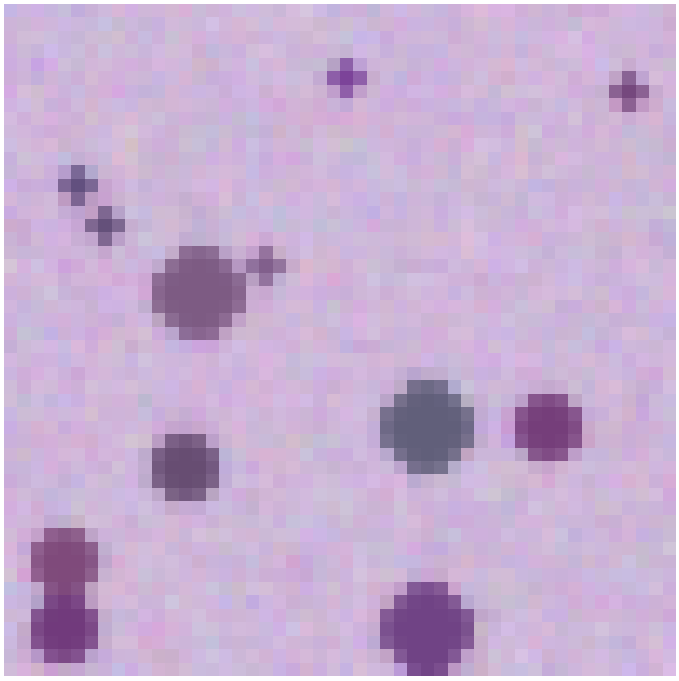
Classification:	**BENIGN**
Confidence Score:	92.2%
Benign Probability:	92.2%
Malignant Probability:	7.8%

Probability Distribution



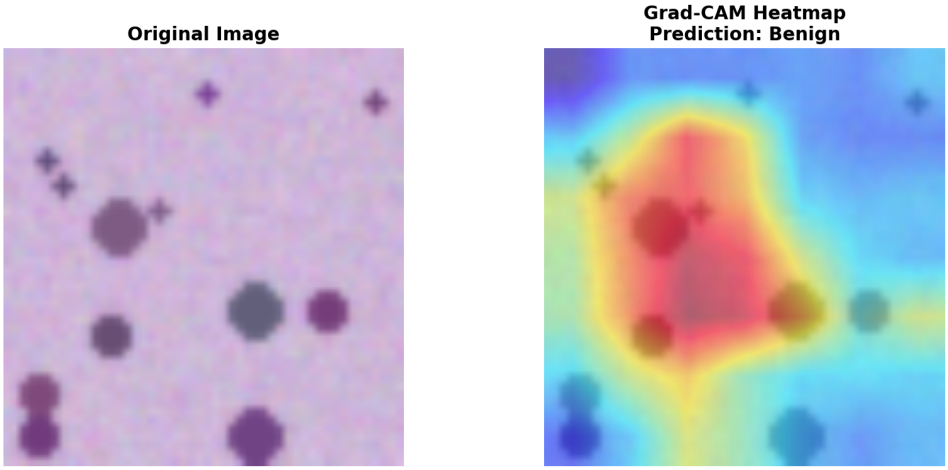
HISTOPATHOLOGY IMAGE ANALYSIS

Original Histopathology Image



Explainability: Grad-CAM Heatmap

The heatmap below highlights the tissue regions that most influenced the AI's decision. Red/yellow areas indicate high importance, while blue areas indicate low importance.



CLINICAL RECOMMENDATIONS

Routine screening after 12 months. Continue regular breast health monitoring. Maintain healthy lifestyle and attend scheduled screenings.

Risk Level:	**LOW**
Follow-up Timeline:	12 months (routine screening)
Specialist Referral:	Primary Care Physician (routine follow-up)

IMPORTANT MEDICAL DISCLAIMER

This report is generated by an AI-assisted diagnostic system and is intended for decision support purposes only.

Key Points:
• This analysis must be validated by qualified medical professionals
• Do NOT make treatment decisions based solely on this AI analysis
• Always consult with board-certified pathologists and oncologists
• This system is not a substitute for professional medical judgment
• False positives and false negatives can occur
• Clinical correlation and additional diagnostic tests are essential

Regulatory Status:
This AI system is for research and educational purposes. It has not been approved by regulatory agencies (FDA, CE, etc.) for clinical diagnostic use.

Professional Review Required:
All findings must be confirmed through standard pathological examination and clinical evaluation by licensed healthcare providers.

TECHNICAL DETAILS

Model Architecture: ResNet50 with Transfer Learning
Training Dataset: Histopathology images (Benign/Malignant classification)
Input Image Size: 224 × 224 pixels
Preprocessing: ImageNet normalization
Explainability Methods: Grad-CAM, SHAP, Integrated Gradients
Software Version: Breast Cancer Detection System v1.0
Processing Timestamp: 2026-02-16 19:26:19

This report was automatically generated by the AI-Assisted Breast Cancer Detection System. For questions or concerns, please consult with your healthcare provider.